Replicating the Effects of Birth Order on Mental Well-Being

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Abstract

Mental health is a significant factor in overall health as it affects the risk for many conditions, and a portion of U.S. adults suffer from mental illness. Known risk factors include adverse childhood experiences, certain medical conditions, and social factors. One of the factors that has not been thoroughly explored is birth order. Past research on birth order has yielded mixed and inconsistent personality effects on personality. Effects on mental well-being are more one-sided, with later-borns experiencing higher mental well-being and prosocial behavior than first-borns. Some researchers hypothesize that later-borns get earlier chances to socialize due to the presence of older siblings, leading to better emotion regulation, thought regulation, and problem-solving. Others note that first-borns may be subjected to more family responsibility and stress. According to intersectionality theories and the multiple jeopardy hypothesis, birth order may also interact with the negative effects of gender or immigration background. This correlational study aims to replicate the effects of birth order on mental well-being and explore which aspects of mental well-being in particular may be affected. The study is expected to fill a gap by recruiting a sample with more women and people of immigrant backgrounds. It also aims to determine whether there is an interaction between birth order, immigration background, and gender on mental well-being. Participants will primarily be recruited from the University of San Francisco General Psychology course. Participants will take a 15-minute online survey that assesses their mental well-being, birth order, gender, and immigrant background.

Background & Significance

The importance of mental health has risen in recent years, especially in developed countries. Mental illness now affects a significant portion of the U.S.—22.8% of adults suffer from mental illness while 17% of youth experience mental health disorders (National Alliance on Mental Illness, 2023). Mental health is a key factor in overall health as it affects the risk for chronic conditions such as diabetes or cardiovascular disease (CDC, 2023). Known risk factors for mental illness include adverse childhood experiences, certain medical conditions, as well as social factors like loneliness and isolation (CDC, 2023). To this date, however, there are still sociocultural factors that have not been thoroughly explored, including birth order, immigration background, and gender.

Birth Order

Effects on Personality

Sulloway (1996) revised and popularized the idea that the mechanism behind different siblings' family roles and personalities is the competition for parental investment. First-borns would be more likely to follow their parents' expectations and values because they received more attention from their parents. They would become more responsible, organized, and academically achieving, which is associated with conscientiousness (Healey & Ellis, 2006). On the other hand, later-borns would struggle with finding their own family role. They would identify less with their parents and older siblings and thus become the unconventional, liberal "rebels" in their family (Healey & Ellis, 2006).

Studies on the effects of birth order and personality are extensive and have yielded mixed results. Black et al. (2018) found that in a sample of Swedish men, earlier-borns were more emotionally stable, socially outgoing, and took more initiative than later-borns. Consequently, first-borns were more likely to work in managerial and leadership positions (Black et al., 2018). The findings of Healy & Ellis (2006) and Dixon et al. (2008) also fit into Sulloway's model, with first-borns exhibiting higher conscientiousness and later-borns displaying more extraversion and openness.

In 1983 however, Ernst & Angst reviewed 40 years of previous research on birth order and echoed the findings of Olneck & Bills (1979): most personality effects were confounded with socioeconomic status and family size—an effect of the between-family designs that were used by many of the researchers. Although some within-family studies did in fact support Sulloway's hypothesis (Sulloway, 2010; Sulloway, 1999; Paulhus et al., 1999; Dixon et al., 2008), the study conducted by Bleske-Rechek & Kelley (2013) utilized a within-family design along with independent self-reports from 69 pairs of both first-born and later-born siblings and found no difference in personality. Parents' reports of their children's personality traits did not differ either (Bleske-Rechek & Kelley, 2013). Finally, a study combining between- and within-family designs found no lasting effects of birth order on personality in a sample of 20,000 people (Rohrer et al., 2015).

Effects on Mental Well-Being

Research on how birth order and these family contexts shape mental well-being is more limited. In a study of 2,886 young adults, mostly male, Fullerton et al. (1989) found that male first-borns scored higher on the negative feelings subscale of psychological well-being than other-borns when controlling for socioeconomic status. This subscale is correlated with psychological distress, including anxiety (Fullerton et al., 1989). Dohrenwend & Dohrenwend (1966) add that in anxiety-arousing situations, first-borns showed more psychiatric symptoms when they are unable to share the experience with others. In a study of 3,744 Japanese children from Tokyo, later-borns exhibited higher prosocial behavior and resilience while displaying fewer total difficulties, conduct problems, and hyperactivity than first-borns (Fukuya et al., 2021). Lawson & Mace (2010) supported this finding in British children—the presence of older siblings predicted higher childhood mental health, while the arrival of younger siblings indicated increased emotional and conduct difficulties. Chandola & Tiwari (2016) achieved similar findings in Indian participants as well: in a comparison between 20 patients with psychiatric symptoms and 20 healthy patients, male first-borns had significantly higher psychiatric morbidity than any other group. First-borns also exhibited statistically significant higher scores in emotional instability compared to last-borns and only children (Chandola & Tiwari, 2016).

Learned Prosocial Behavior in Younger Siblings

One possible mechanism behind the effect of birth order on mental well-being is that older siblings provide social benefits to younger ones (Lawson & Mace, 2010; Fukuya et al., 2021). Since children spend more time with their siblings than with their parents in daily life, early sibling interactions promote thought regulation and understanding of emotions. Conflicts during play promote prosocial behaviors like problem-solving, teaching, sharing, and cooperation. In other words, later-borns have more opportunities for in-depth social development from an early age.

Family Responsibility, Pressure, and Stress for Older Siblings

Another explanation is that younger siblings benefit from having older siblings as a secure base of attachment or complementary caretakers (Fukuya et al., 2021). These older siblings, however, may suffer from their position in the family. As the oldest child, they may be subject to increased parental pressure and responsibility. Conduct problems can also arise in response to the stress of a changing home environment—per Adler’s dethronement theory, first-borns may feel jealousy after receiving less parental attention and interactions than the later-borns (McHale et al., 2012).

Gender. The negative effects of being a first-born could be more pronounced for women as they report about twice as much depressive and anxious symptomatology compared to men (Johns Hopkins Medicine, 2024) and may also score higher on neuroticism (Eysenck & Eysenck, 1975; Viken et al., 1994), a predictor of poorer mental well-being. In a similar fashion to first-borns, women are persistently under pressure to cater to domestic labor and other family responsibilities that can generate significant amounts of stress, especially in conjunction with work commitments (Austen & Birch, 2000; Somashekher, 2018).

Immigration Backgrounds & Mental Health

With the increasing rate of globalization, there will be more immigrants and families with immigrant backgrounds. The American Immigration Council (2021) finds that 13.6% of U.S. residents are foreign-born and the United Nations Regional Information Centre for Western Europe (2023) estimates that 5.3% of people in the European Union are non-EU citizens. Moving to a new country often involves bureaucratic whirlwinds, language barriers, cultural differences, raising families, and economic difficulty. These effects can have a long-lasting impact on subsequent generations. Twenty-seven percent of people aged 15-34 in developed nations have a migrant background (Organisation for Economic Cooperation and Development, 2018).

Current research on immigrant mental health is limited. Counter-intuitively, many researchers have found a so-called healthy migrant effect: many foreign-born populations have better health than native-born populations (Ali, 2002; McKay et al., 2003; Perez, 2002). This finding seems to diminish over time, however, and is less established for mental health (Elshahat et al., 2022). A possible explanation for the healthy migrant effect may be selection bias—those who are able to immigrate might already be pre-selected to be more resilient (Lee, 1996).

Nevertheless, migrating and adjusting to a new country places significant stress on people, particularly for women (Furnham & Bochner, 1986; Furnham & Shiekh, 1993). Another possibility is that many first-generation immigrants come from cultures that hold a stigma against mental health issues, so mental health challenges might be interpreted under somatic symptoms and go underreported (Millet et al., 2022).

The immigrant paradox describes the issue of mental health in second- and third-generation immigrants in comparison to the first (parent) generation. Montazer & Wheaton (2011) demonstrated that in immigrant families of low economic development, mental health problems increased across generations. In a review of immigration and mental health, Alegria et al. (2017) found that U.S.-born Asian-Americans had higher psychiatric rates than foreign-born Asians. The same difference was discovered between U.S.-born and foreign-born Latinos (Alegria et al, 2017). These findings became less clear and more inconclusive when broken down by ethnic subgroup (Alegria et al., 2017).

Cultural Factors

Many immigrants come from cultures that may be more vulnerable to mental health problems. Kim et al. (2019) reviewed culture-specific factors that affect the mental health of children of immigrants and found that Chinese-American parents stress academic achievement at the expense of adolescent mental health (Kim et al., 2015). These expectations and results may differ depending on birth order: in Japanese culture, first-borns are required to behave more mature and responsible while taking care of younger siblings (Holloway & Behrens, 2002), resulting in a stricter upbringing for first-borns (Power et al., 1992), which leads to poorer mental well-being (Fukuya et al., 2021).

Specific Aims

Aspects Of Mental Well-Being

While birth order may have mixed effects on personality, it could be a predictor of higher mental health in later-borns through social benefits and lower mental health in first-borns through parental pressure and dethronement. Previous studies, however, have not yet explored the specific aspects of mental well-being that are affected by birth order. Research has shown that people affected by psychological disorders experience different impacts on well-being (Honey et al., 2015; Goodman et al., 2019). For example, people with depression respond less strongly to negative events than healthy people while people with social anxiety disorder feel persistently low levels of positive emotions in both social and non-social contexts (Goodman et al., 2019). Oftentimes, trauma survivors lead purposeful, appreciative lives upon overcoming and coping with their traumatic or stressful experiences (Goodman et al., 2019). These differences are factors in designing effective interventions for people with different conditions (Goodman et al., 2019). Similarly, it would be helpful to explore how different aspects of mental well-being may be impacted by birth order.

Intersectionality

One common theme that has emerged in this literature review is that being a first-born, woman, or having an immigrant background can all be associated with some form of increased family pressure. These pressures can emerge in the form of domestic labor, academic achievement, or caretaking, which then have a detrimental effect on well-being (Fukuya et al., 2021; Austen & Birch, 2000; Kim et al., 2015). The colloquial term “eldest [immigrant] daughter syndrome” has also been coined to summarize the challenging family context of the increased responsibilities and stressors for young girls (Tricaso, 2023). It is possible that gender and immigrant background interact and moderate the effect of birth order on mental well-being; the effects might be amplified under certain conditions. Intersectionality theories explain how a person's multiple social identities interact and lead to opportunities and oppressions that are not necessarily brought about by the separate social identities on their own (Taylor, 2019). For instance, black women face forms of unique discrimination that are not experienced by white women or black men (United Way of the National Capital Area, 2023). In line with this framework, the multiple jeopardy hypothesis predicts that the negative effect of being a member of multiple marginalized social groups is higher than the additive negative effect of all those social groups (King, 1998). Kern et al. (2020) examined the relationship between low socioeconomic status (SES), immigrant background, and gender to observe their effects on mental well-being. In the sample of over 100,000 adolescents from various European countries, the negative effects of low SES, immigrant background, and being a girl were additive and not interactive (Kern et al., 2020).

This Study

Current research on the effects of birth order on mental well-being has expressed the need for a larger sample of women and people with immigrant backgrounds. Additionally, previous studies that found birth order effects focused on children (Fukuya et al., 2021; Lawson & Mace, 2010) or were conducted several decades ago (Fullerton et al., 1989), so there is a need to assess whether those birth order effects have longer-lasting effects on adults in a modern context.

My current study aims to bridge some of the gaps by exploring the effects of birth order on different aspects of mental well-being in adults. I will also test the multiple jeopardy hypothesis with the factors of birth order, immigration background, and gender on mental well-being. Here are the specific aims and hypotheses:

(1) Replicate the negative effects of birth order on aspects of mental well-being in adults, particularly in a sample with more women and people with immigrant backgrounds.

H1: Overall, first-borns will report lower mental well-being than later-borns.

H2: First-borns with immigrant backgrounds will report lower mental well-being than later-borns with immigrant backgrounds.

(2) Explore how birth order affects the different aspects of mental well-being.

(3) Test the multiple jeopardy hypothesis with the factors birth order, immigrant background, and gender on mental well-being.

H3: Immigration background and gender moderate the negative effect of birth order on mental well-being; first-born daughters with immigrant backgrounds will report the lowest mental well-being out of all groups.

Methods

Participants

The sample will consist of students at the University of San Francisco (USF). Most participants will be students enrolled in the General Psychology course in the Fall of 2024. Other participants will be recruited from flyers publicized through student organizations. The target sample size is 100 participants based on the size of the sample pool. The sample is expected to be representative of the USF undergraduate population, primarily aged 18-30, of which 64% identify as female and 36% identify as male. The sample is expected to include a diverse range of racial, ethnic, and socioeconomic backgrounds. Of the current student body, 7% identify as African American/Black, 26% identify as Asian/Asian American, 24% identify as Hispanic/Latino, 24% identify as White, and 18% identify as multicultural or with a different group.

Students in the General Psychology course are given 30 minutes of class credit for participating in research studies at USF. Alternatively, students can opt out of the assignment by writing a research report instead.

Design & Materials

Participants complete the study in the form of an online Qualtrics survey. First, participants are given a consent form outlining the potential risks, benefits, confidentiality, compensation, and broad overview of the study. After completing the survey, participants are debriefed and given course credit. The study is expected to take 15 minutes in total. The following sections describe the scales used in the order in which they will appear on the survey.

Ryff Psychological Well-Being Scale

The 42-item Ryff Psychological Well-Being Scale is a construct of well-being and happiness (Ryff, 1989). Participants rate how strongly they agree or disagree with 42 statements on a 7-point Likert scale. The Psychological Well-Being Scale consists of six different subscales and produces subscores for each: autonomy (e.g., “I have confidence in my opinions, even if they are contrary to the general consensus”), environmental mastery (e.g., “In general, I feel I am in charge of the situation in which I live”), personal growth (e.g., “I think it is important to have new experiences that challenge how you think about yourself and the world”), positive relations with others (e.g., “People would describe me as a giving person, willing to share my time with others”), purpose in life (e.g., “Some people wander aimlessly through life, but I am not one of them”), and self-acceptance (e.g., “When I look at the story of my life, I am pleased with how things have turned out”). The scale has been used with U.S. adults of all ages, including students (Fatemeh & Valiolah, 2012) and people from lower-income backgrounds (Ryff & Keyes, 1995; Curhan et al., 2014). The scale has also been validated in a diverse range of non-Western populations, including Asian Americans, Asian international students, and Latino students (Iwamoto & Liu, 2010; Bahamon et al., 2019). Thus, this scale is expected to be a valid and reliable construct of mental well-being for this study.

Patient Health Questionnaire (PHQ-9)

The PHQ-9 is a nine-item questionnaire used to screen for the presence and severity of depression (Kroenke et al., 2001). Participants are asked to respond how often they have experienced certain relevant symptoms over the past two weeks (e.g., “Little interest or pleasure in doing things”, “Trouble concentrating on things, such as reading the newspaper or watching television”). Question nine, which assesses suicidal ideation, was taken out due to concerns about mandated reporting. The PHQ-9 has been tested for its reliability and validity and is used for both clinical and research purposes (Kroenke et al., 2001).

Generalized Anxiety Disorder Scale (GAD-7)

The GAD-7 is a seven-item questionnaire that assesses presence and severity of generalized anxiety disorder (Spitzer et al., 2006). Participants are asked to respond how often they have experienced certain relevant symptoms over the past two weeks (e.g., “Feeling nervous, anxious, or on edge”, “Trouble relaxing”). The GAD-7 has been tested for its reliability and validity in the general population (Loewe et al., 2008).

Demographics Questionnaire, Immigrant Background, and Birth Order

The demographics questionnaire consists of 10-12 questions, depending on certain question responses. Questions first ask about participants’ age, sex, gender, race, and ethnicity. To assess and control for socioeconomic status, participants are asked about their parents’ or guardians’ education level because college students have not yet finished their education and are more reliant on their parents’ or guardians’ income. To determine immigrant background, participants are asked whether they themself, one of their parents, or one of their grandparents was born in a foreign country. Birth order is determined by whether a participant identifies as the only child, first-born, a middle-born, or last-born. For exploratory analyses, participants are also asked how many younger and older siblings they have, if applicable.

Data Analysis Plan

All statistical analyses will be conducted on Jamovi. To account for the multiple comparisons of mental well-being, I have applied a Bonferroni correction to adjust the alpha level to .01667. To test the effect of birth order on mental well-being (Hypothesis 1), I will conduct a t-test to compare mental well-being between first-borns and later-borns. For all analyses involving birth order, middle-borns and last-borns will be aggregated into a single group—later-borns. Only children will be excluded from the main analyses. To control for socioeconomic status, I will be conducting an ANCOVA. To evaluate the effect of birth order on mental well-being in people with immigrant backgrounds (Hypothesis 2), I will conduct the same t-test on first-borns and later-borns with immigrant backgrounds, respectively. For all analyses involving immigrant background, participants will be grouped into those who do not have an applicable immigration background (no parent or grandparent was born in a foreign country) and those who do. Finally, I will use an ANOVA to evaluate whether there is an interaction between birth order, gender, and immigration background on the effect on mental well-being (Hypothesis 3). If a statistically significant interaction is found, Tukey's post hoc test will be used to determine the direction and size of the interaction effect. To control for socioeconomic status, I will conduct an ANCOVA.

Potential Challenges

There are a few potential challenges that participants may face. Some additional demographic questions ask participants about additional aspects of their identity, such as which race/ethnicity they are perceived as in public and whether they identify as nonbinary/transgender, etc. Participants may experience discomfort disclosing certain identities. To alleviate discomfort associated with the demographic questionnaire, this section of the survey is programmed to force a response to only the questions about gender, socioeconomic status, immigration background, and birth order as they are my independent variables and covariates. The other questions merely request a response. Questions in the Ryff Psychological Well-Being Scale may cause participants to reflect on uncomfortable memories or challenges that they may be facing with their identity.

Similarly, the PHQ-9 and GAD-7 may remind participants of emotional struggles that they are experiencing. Additional potential risks include boredom and fatigue while filling out the survey.

To address these concerns, the consent form reminds participants that those who sign up and arrive on time will receive credit regardless of whether they decide to withdraw consent later and discontinue participation in the study.

Potential challenges regarding data collection and analysis include the statistical power needed to observe an interaction effect when testing Hypothesis 3.

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